A MISSING TERM IN AN EXPLICIT FORMULA FOR GOLDBACH AVERAGES

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Abstract. Let $r_G(n) = \sum_{m_1+m_2=n} \Lambda(m_1)\Lambda(m_2)$. Cantarini gave an explicit formula for $\sum'_{n\le 2N} r_G(n)$ whose main new term is a double sum over pairs of nontrivial zeros of ζ, with remainder $F(N) = O(N)$. We show that the bookkeeping of the proof drops the term $\big(\sum_\rho N^\rho/\rho\big)^2$. With this term restored the formula holds with remainder $O(N)$; without it, Littlewood's oscillation theorem gives $\limsup F(N)/N = +\infty$, so the formula as stated is false. Using the corrected formula and the pointwise lower bound of Bhowmik and Schlage-Puchta, we show that the double sum is not $O(N)$, with no hypothesis on the zeros. Both results rely on two lemmas of Cantarini's paper, which we do not re-prove. Hence the estimate in Cantarini's Conjecture 2 fails; since the conjecture is stated under the Riemann hypothesis, it is equivalent to the failure of the Riemann hypothesis. Two independent implementations, at 1969 values of N up to $6 \cdot 10^6$, confirm the corrected formula.

1. Introduction

Let Λ be the von Mangoldt function, $\psi(x) = \sum_{n\le x} \Lambda(n)$, and

$$r_G(n) = \sum_{\substack{m_1,m_2\le n\\ m_1+m_2=n}} \Lambda(m_1)\Lambda(m_2).$$

The average of r_G has a long history. Fujii [4] obtained the first two terms of an asymptotic expansion of $\sum_{n\le N} r_G(n)$, with error $O\big((N\log N)^{4/3}\big)$; Goldston and Suriajaya [6] revisited Fujii's formula. Cesàro-weighted averages $\frac{1}{\Gamma(k+1)}\sum_{n\le N} r_G(n)(N-n)^k$, $k>0$, admit explicit formulae by the work of Brüdern, Kaczorowski and Perelli [2]. Cantarini [3] treated the unweighted case $k=0$.

1.1. **Cantarini's formula.** Write $\sum'_{n\le 2N} r_G(n) = \sum_{n\le 2N} r_G(n) - \frac12 r_G(2N) =: S'(N)$. The sums over zeros below are symmetric, $\sum_\rho = \lim_{T\to\infty}\sum_{|\gamma|\le T}$. Let $B_z(a,b)$ be the incomplete Beta function and put

$$T(N) = 2\sum_{\rho_1}(2N)^{\rho_1}\Gamma(\rho_1)\left[\sum_{\rho_2}\frac{(2N)^{\rho_2}\Gamma(\rho_2)}{\Gamma(\rho_1+\rho_2+1)} - \sum_{\rho_2}\frac{(2N)^{\rho_2}}{\rho_2}\big(B_{1/N}(\rho_2+1,\rho_1)+B_{1/2}(\rho_1,\rho_2+1)\big)\right]. \tag{1.1}$$

Theorem 1 of [3] states that for every integer $N>2$

$$S'(N) = 2N^2 - 2\sum_\rho \frac{(2N-2)^{\rho+1}}{\rho(\rho+1)} + T(N) + F(N), \qquad F(N) = O(N), \tag{1.2}$$

"without the assumption of the Riemann hypothesis". Conjecture 2 of [3] reads: *Under RH we have* $T(N)/2 \ll N$ (in [3] the factor 2 of (1.1) is not part of the conjectured quantity). Theorem 3 of [3] states: *assume that Conjecture 2 holds; then every interval* $[2N, 2N+2H]$ *with* $H = C\log\log N$ *and* $C>0$ *sufficiently large contains a Goldbach number.*

The analysis below refers to the arXiv version of [3] (arXiv:1801.08475v1); the published text (Indian J. Math. 61 (2019) 253–279) could not be consulted. All equation and page numbers of [3] quoted here refer to the arXiv version.

1.2. **Results.** Put $c_0 = \frac{\zeta'}{\zeta}(0) = \log 2\pi$ and

$$P(x) = \sum_\rho \frac{x^\rho}{\rho} = x - \psi_0(x) - \log 2\pi - \tfrac12\log(1-x^{-2}), \qquad x>1, \tag{1.3}$$

where $\psi_0(x) = \psi(x) - \frac12\Lambda(x)$ at integers (von Mangoldt's explicit formula).

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Theorem 1.1 (Corrected form of [3], Theorem 1)**.** *Assume Lemmas 5 and 6 of* [3]*. For every integer $N > 2$,*

$$S'(N) = 2N^2 - 2\sum_{\rho} \frac{(2N-2)^{\rho+1}}{\rho(\rho+1)} + T(N) + \Big(\sum_{\rho} \frac{N^{\rho}}{\rho}\Big)^2 + \widetilde{F}(N), \qquad \widetilde{F}(N) = O(N),$$

with no hypothesis on the zeros of ζ.

Corollary 1.2. *With $F(N)$ defined by* (1.2)*, $F(N) = P(N)^2 + \widetilde{F}(N)$ and $\limsup_{N\to\infty} F(N)/N = +\infty$. In particular* [3]*, Theorem 1, is false as stated.*

Theorem 1.3. *Assume Lemmas 5 and 6 of* [3] (*hence Theorem 1.1*)*. There is $C > 0$ such that, for all sufficiently large N,*

$$T(N+1) - T(N) \;\geq\; \tfrac{1}{2}\big(r_G(2N) + r_G(2N+2)\big) - CN.$$

Consequently $T(N) \neq O(N)$. No hypothesis on the zeros of ζ is used.

Corollary 1.4. *Under the assumption of Theorem 1.3, the estimate $T(N)/2 \ll N$ of* [3]*, Conjecture 2, is false. Hence Conjecture 2, as stated ("Under RH . . . "), is equivalent to the failure of RH; in particular it is false if RH holds.*

"Unconditional" in Theorem 1.3 means "without RH"; it does not mean "independently of [3]": Theorems 1.1 and 1.3 rest on Lemmas 5 and 6 of [3], which we have not re-proved (Section 2.4). The consequences for [3], Theorem 3, are discussed in Section 4.3. Theorem 1.1 alone says nothing about the size of $T(N)$; Theorem 1.3 needs, in addition, the pointwise lower bound for r_G of Bhowmik and Schlage-Puchta [1].

2. The explicit formula and the missing term

Let $\psi_1(x) = \int_0^x \psi(t)\,dt$ and

$$A(x) = \sum_{\rho} \frac{x^{\rho+1}}{\rho(\rho+1)} = \frac{x^2}{2} - \psi_1(x) - c_0 x + \frac{\zeta'}{\zeta}(-1) - \sum_{r\geq 1} \frac{x^{1-2r}}{2r(2r-1)},$$

the second expression coming from the explicit formula for ψ_1 ([7], Theorem 28). Then $A'(x) = P(x)$ for almost all x. Put $\Phi(N) = S'(N) - 2N^2 + 2A(2N-2)$, so that Theorem 1.1 reads $\Phi(N) = T(N) + P(N)^2 + \widetilde{F}(N)$.

2.1. **Starting point.**

The proof in [3] starts from eq. (18) of [3] (p. 10),

$$\sum_{n\leq 2N} r_G(n) = 2\sum_{n\leq N} \Lambda(n)\,\psi(2N-n) - \psi(N)^2, \tag{2.1}$$

obtained by splitting the triangle $\{i + j \leq 2N\}$ into a square and two triangles. Note the coefficient -1 of $\psi(N)^2$.

For every integer $m \geq 2$, (1.3) gives $\psi(m) = m - P(m) - c_0 - \frac{1}{2}\log(1-m^{-2}) + \frac{1}{2}\Lambda(m)$. Inserting this with $m = 2N - n$ into (2.1), the part $\frac{1}{2}\Lambda(2N-n)$ contributes

$$2\sum_{n\leq N} \Lambda(n)\,\tfrac{1}{2}\Lambda(2N-n) = \tfrac{1}{2}r_G(2N) + \tfrac{1}{2}\Lambda(N)^2.$$

The first term is absorbed by $\sum'$; the second is $O(\log^2 N)$. ([3], §3.5, writes $\frac{1}{2}r_G(2N)$ for this sum; the difference $\frac{1}{2}\Lambda(N)^2$ is harmless.)

2.2. **Term-by-term bookkeeping.**

Write also $\psi(N) = N - P(N) - c_0 + \frac{1}{2}\Lambda(N) + O(N^{-2})$, and abbreviate $P = P(N)$, $\Lambda = \Lambda(N)$. The remaining contributions to (2.1) are collected in Table 1. Block (ii) is obtained in [3], §3.2, from Lemma 6 of [3] with $x = N$, $y = 2N$, $\alpha = \rho$; its result is eqs. (33)–(36) of [3], where (33) contains explicitly $+2\sum_{\rho_1}\sum_{\rho_2} \frac{N^{\rho_1+\rho_2}}{\rho_1\rho_2} - \Lambda(N)\sum_{\rho}\frac{N^{\rho}}{\rho}$, and (34)–(35) form $T(N)$. The term $+2P^2$ comes from eq. (27). In block (ii) the terms $-4P(2N-2)$, $2c_0P(2N-2)$ and $\log\frac{3}{4}\,P(2N-2)$ are $o(N)$ by eq. (31) of [3], and the term ω of Lemma 6 is $\ll Ne^{-c\sqrt{\log N}}$ by eq. (32).

TABLE 1. Contributions to (2.1), with $P = P(N)$, $\Lambda = \Lambda(N)$.

Block	Origin in [3]	Contribution
(i) $2\sum_{n\le N}\Lambda(n)(2N-n)$ $= 2N\psi(N) + 2\psi_1(N)$	§3.1, (20)–(22)	$2N\psi(N) = 2N^2 - 2NP + N\Lambda - 2c_0N + O(1/N)$ $2\psi_1(N) = N^2 - 2A(N) - 2c_0N + O(1)$
(ii) $-2\sum_{n\le N}\Lambda(n)P(2N-n)$	§3.2, (24)–(36)	$-2A(2N-2) + 2A(N) + T(N) + \mathbf{2P^2} - \Lambda P + O\big(Ne^{-c\sqrt{\log N}}\big)$
(iii) $-2c_0\psi(N)$	§3.3	$-2c_0N + 2c_0P + O(\log N)$
(iv) logarithmic term	§3.4	$O(1)$
(v) $-\psi(N)^2$	(18)	$-N^2 + 2NP - N\Lambda - \mathbf{P^2} + P\Lambda + 2c_0N - 2c_0P + O(\log^2 N)$

Proof of Theorem 1.1. Add the rows of Table 1. The terms $\pm 2NP$ cancel between (i) and (v), $\pm N\Lambda$ between (i) and (v), $\pm 2A(N)$ between (i) and (ii), $\pm\Lambda P$ between (ii) and (v), and $\pm 2c_0P$ between (iii) and (v). The coefficient of N^2 is $2 + 1 - 1 = 2$. What remains is

$$S'(N) = 2N^2 - 2A(2N-2) + T(N) + (2-1)\,P(N)^2 + \big(-2c_0 - 2c_0 - 2c_0 + 2c_0\big)N + o(N),$$

where the coefficient $2 - 1$ of $P(N)^2$ comes from (ii) and (v), and the four contributions $-2c_0N$, $-2c_0N$, $-2c_0N$, $+2c_0N$ come from $2N\psi(N)$, $2\psi_1(N)$, (iii) and (v). □

Remark 2.1. The computation gives the more precise statement $\widetilde{F}(N) = -4\log(2\pi)\,N + o(N)$, with $-4\log 2\pi = -7.351508\ldots$; this is matched to six decimal places by the numerics of Section 5.

2.3. **Where the term disappears.** In [3], §3.6 (p. 14), the proof concludes: "Expanding $\psi^2(N)$ in (18) with its explicit formula and observing that some terms cancel each other out (see, for example, (23) and (24)) we get that . . . ", followed by the statement of Theorem 1. This is where $P(N)^2$ is lost: $+2P(N)^2$ from (27)/(33) and $-P(N)^2$ from $-\psi(N)^2$ in (2.1) leave $+P(N)^2$, not 0.

The truncated version of Theorem 1 in [3] (p. 2) carries the same signature. It contains

$$+2\sum_{|\gamma_1|\le T'}\frac{N^{\rho_1}}{\rho_1}\sum_{|\gamma_2|\le T''}\frac{N^{\rho_2}}{\rho_2} - 2\Big(\sum_{|\gamma_1|\le T'}\frac{N^{\rho_1}}{\rho_1}\Big)^2.$$

The first term comes from the truncated Lemma 6 (eq. (43), p. 16) and agrees with (27); the square comes from $-\psi(N)^2$, expanded in §3.13 with the same cutoff T', and its coefficient should be -1, not -2. With -2 the two terms tend to $2P^2 - 2P^2 = 0$ as $T', T'' \to \infty$, which explains why nothing appears in the untruncated statement. With the correct coefficient the expression tends to $+P(N)^2$, so the claim "$F(N, T', T'') \ll N$ uniformly in T', T''" fails as well. Both versions thus reflect a single slip of coefficient.

2.4. **What has not been re-verified.**

- Lemma 5 of [3], quoted from Ramaré–Saouter [11], Lemma 4, including its extension to complex g with $\int |g| < \infty$ and the interchange of sum and integral for $g(t) = (2N-t)^{\rho_1 - 1}$.
- Lemma 6 of [3] (pp. 5–9): its application (24)–(28) was checked, its proof was not rewritten independently.
- The interchange of order in the double series ([3], eqs. (29)–(30)); see also the comment on prime powers in Section 5.2.
- The bound $\widetilde{F}(N) = O(N)$ is an assembly of the estimates (31), (32), (36) and §§3.3–3.4 of [3], checked but not re-proved from scratch.

3. $F(N)$ IS NOT $O(N)$

Proof of Corollary 1.2. By Theorem 1.1 and (1.2), $F(N) = P(N)^2 + \widetilde{F}(N)$ with $|\widetilde{F}(N)| \le C_1 N$. By Littlewood [8] (see [7], Chapter V, Theorem 35, and [9], Theorem 15.11), $\psi(x) - x = \Omega_\pm\big(x^{1/2}\log\log\log x\big)$ unconditionally.

Suppose $\psi(x) - x \le -c\sqrt{x}\log\log\log x$ with $x \in [N, N+1)$. Then $\psi(N) = \psi(x)$ and $N \le x$, so $N - \psi_0(N) \ge x - 1 - \psi(x) \ge c\sqrt{x}\log\log\log x - 1$, and by (1.3) $P(N) \ge c'\sqrt{N}\log\log\log N$. The case Ω_+ is analogous; the sign is irrelevant because P enters squared. Hence, for infinitely many N,

$$\frac{F(N)}{N} \;\ge\; c'^2(\log\log\log N)^2 - C_1 \;\longrightarrow\; \infty.$$ □

Remark 3.1. Schmidt's $\Omega(\sqrt{x})$ would only give $P(N)^2 \geq cN$, compatible with $F = O(N)$; the factor tending to infinity in Littlewood's theorem is needed. If RH fails, $\psi(x) - x = \Omega(x^{\Theta-\varepsilon})$ with $\Theta > \frac{1}{2}$ and the conclusion is stronger. In the other direction, $F(N) \ll N \log^4 N$ under RH, since $P(N) \ll \sqrt{N} \log^2 N$, and $F(N) \ll N^2 \exp(-c\sqrt{\log N})$ without RH.

4. Conjecture 2

4.1. **A pointwise lower bound.**

Lemma 4.1 (Bhowmik–Schlage-Puchta [1])**.** $r_G(n) = \Omega(n \log\log n)$*, unconditionally.*

This is proved inside the proof of the Ω-part of Theorem 1.1 of [1], using Gallagher's density estimate [5], Theorem 7, and a treatment of a possible Siegel zero; it ends with $\max_{n\leq 4x} G(n) \gg x/(2\varphi(q)) \gg x \log\log x$, where $G = r_G$. We stress that the Ω-result for the *summatory* function in Theorem 1.1 of [1] is conditional on RH and is not what is used here.

For Theorem 1.3 the rate $\log\log n$ is not needed. The argument of [1] with a fixed modulus, which only needs the prime number theorem in arithmetic progressions, gives the following.

Lemma 4.2. $\limsup_{n\to\infty} r_G(n)/n = +\infty$*, the limit superior being attained along even n.*

Proof. Let q be a primorial, so $2 \mid q$. For $(a, q) = 1$, the pairs (m_1, m_2) with $m_1 \equiv a$, $m_2 \equiv -a \pmod q$ and $m_1, m_2 \leq x$ satisfy $q \mid m_1 + m_2 \leq 2x$; pairs from distinct classes a are distinct, and $r_G \geq 0$. Hence, by the prime number theorem in progressions to the fixed modulus q,

$$\sum_{\substack{n\leq 2x \\ q\mid n}} r_G(n) \;\geq\; \sum_{(a,q)=1} \psi(x; q, a)\, \psi(x; q, -a) = (1 + o(1))\frac{x^2}{\varphi(q)} \qquad (x \to \infty).$$

There are at most $2x/q$ terms on the left, so some $n = n(x) \leq 2x$ with $q \mid n$ satisfies $r_G(n) \geq (1 + o(1))\frac{q}{2\varphi(q)}x \geq (1 + o(1))\frac{q}{4\varphi(q)}n$. Since $r_G(n(x)) \gg_q x$, $n(x) \to \infty$. Thus $\limsup r_G(n)/n \geq q/(4\varphi(q))$ for every primorial q, and $q/\varphi(q) = \prod_{p\mid q}(1 - 1/p)^{-1} \to \infty$. All $n(x)$ are even. □

4.2. **Proof of Theorem 1.3.**

We use only that $\Phi(N) = T(N) + E(N)$ with $E(N + 1) - E(N) = O(N)$. By Theorem 1.1, $E = P^2 + \widetilde{F}$. Since $|P(N + 1) - P(N)| \leq 1 + \log(N + 1) + O(N^{-2})$ and, by the prime number theorem with the classical zero-free region ([3], eq. (31)), $|P(N + 1) + P(N)| \leq 2 \sup_{t\leq N+1} |P(t)| \ll Ne^{-c\sqrt{\log N}}$, we get $P(N + 1)^2 - P(N)^2 \ll Ne^{-c\sqrt{\log N}} \log N = o(N)$ (and $\ll \sqrt{N} \log^3 N$ under RH). Hence $E(N + 1) - E(N) = O(N)$.

Step 1. By the definition of $\sum'$,

$$S'(N + 1) - S'(N) = \tfrac{1}{2}r_G(2N) + r_G(2N + 1) + \tfrac{1}{2}r_G(2N + 2). \tag{4.1}$$

The half weights at the two ends add up; they do not cancel.

Step 2. $2(N + 1)^2 - 2N^2 = 4N + 2$, and $A(2N) - A(2N - 2) = \int_{2N-2}^{2N} P(t)\, dt$, so $|A(2N) - A(2N - 2)| \leq 2 \sup |P| = o(N)$.

Step 3. Therefore, using $r_G(2N + 1) \geq 0$,

$$\begin{aligned} T(N + 1) - T(N) &= \tfrac{1}{2}r_G(2N) + r_G(2N + 1) + \tfrac{1}{2}r_G(2N + 2) - (4N + 2) + o(N) - \big[E(N + 1) - E(N)\big] \\ &\geq \tfrac{1}{2}\big(r_G(2N) + r_G(2N + 2)\big) - C_2N. \end{aligned}$$

Step 4. If $|T(N)| \leq C_3N$ for $N \geq N_0$, then $r_G(2N) \leq 2(C_2 + 2C_3 + 1)N$ for all large N, i.e. $r_G(n) \ll n$ for even n, contradicting Lemma 4.2 (or Lemma 4.1). Hence $T(N) \neq O(N)$. Only the classical zero-free region and Lemma 4.2 were used, not RH. □

Remark 4.3. (a) The midpoint convention plays no role: with $\sum$ instead of $\sum'$, (4.1) becomes $r_G(2N + 1) + r_G(2N + 2)$ and the argument is unchanged. (b) The argument works equally well with (1.2) as stated ($E = F = O(N)$), so it is robust to the coefficient slip of Section 2; it is not robust to an error in Lemmas 5–6 of [3] or in the assembly of $\widetilde{F} = O(N)$. (c) In the proof of Theorem 3 of [3] (p. 3), the identity written "for every $H > 0$" is, for $H = 1$, already $r_G(2N + 1) + \frac{1}{2}r_G(2N + 2) + \frac{1}{2}r_G(2N) = 4N + 2 + O(N)$, which forces $r_G(M) \ll M$ on even M; only the upper bound $r_G(M) \ll M \log\log M$ ([9], Theorem 3.13, as quoted in [3]) is used there.

4.3. **Two readings of Conjecture 2 and Theorem 3.**

Proof of Corollary 1.4. Write E for the estimate $T(N)/2 \ll N$. By Theorem 1.3, E is false. Read literally, Conjecture 2 is the implication RH $\Rightarrow E$, which is therefore equivalent to ¬RH: false if RH holds, vacuously true if RH fails. □

For Theorem 3 of [3] two readings must be distinguished.

(a) *Literal reading* ("Assume that Conjecture 2 holds"). The hypothesis holds exactly when RH fails, and Theorem 3 then asserts: *if RH fails, every interval* $[2N, 2N + 2C\log\log N]$ *contains a Goldbach number.* This is not vacuous; it is *not proved*, because the proof on p. 3 of [3] uses the estimate E itself, which is false.

(b) *Reading the hypothesis as the estimate E.* Then Theorem 3 is vacuous, since its hypothesis never holds.

In short: the estimate in Conjecture 2 fails unconditionally; Conjecture 2, as stated, is equivalent to the failure of RH; and the hypothesis actually used in the proof of Theorem 3 never holds.

5. Numerical confirmation

5.1. **Method.** We computed $F(N)$ exactly, without truncating any sum over zeros, in two independent implementations (B does not import code from A):

- a sieve for Λ (numpy, double precision; sums with `math.fsum` or extended precision), no number-theory library;
- $S'(N)$ by (2.1) (A) and directly as $\sum_m \Lambda(m)\psi(2N-m)$ (B);
- $A(x)$ by the explicit formula for ψ_1 (B includes $\frac{\zeta'}{\zeta}(-1)$ and the tail $\sum_r x^{1-2r}$; A drops the $O(1)$);
- $T(N)$ by the representation

$$T(N) = 2\int_N^{2N-2} P(2N-s)\,dP(s) = 2\big(A(N) - A(2)\big) - 2\sum_{N\le m\le 2N-2}{}' \Lambda(m)P(2N-m) + O(1/N), \tag{5.1}$$

with weight $\frac{1}{2}$ at the endpoints, derived from Lemma 5 of [3] and the identity, for each pair (ρ_1, ρ_2),

$$\Gamma(\rho_1)\frac{(2N)^{\rho_2}\Gamma(\rho_2)}{\Gamma(\rho_1+\rho_2+1)} - \frac{(2N)^{\rho_2}}{\rho_2}\big(B_{1/N}(\rho_2+1,\rho_1) + B_{1/2}(\rho_1,\rho_2+1)\big) = \frac{(2N)^{\rho_2}}{\rho_2}\int_{1/N}^{1/2} u^{\rho_2}(1-u)^{\rho_1-1}\,du,$$

checked with mpmath at 30 digits (relative error $\le 4.1\cdot 10^{-29}$).

Since an integration by parts can change (5.1) by $\pm 2P^2$ according to the boundary convention, (5.1) was validated against the zeros rather than assumed (Section 5.2). Summing the double series directly for $N \sim 10^6$ is not an option: the truncation error $O(N^2\log^2/T')$ of [3] is $\gg N$ with 10^4–10^5 zeros.

Samples: (A) 1500 random $N \in [10^6, 5\cdot 10^6]$ plus 60 values with $2N$ a multiple of 30030, 510510 or 9699690 and their neighbours (1560 values); (B) 280 random $N \in [2\cdot 10^5, 6\cdot 10^6]$ plus values with $2N$ a multiple of 570570, 510510 or $2\cdot 4849845$ and neighbours (409 values, $N \in [221121, 5990986]$). All runs used a CPU with at most four threads and no GPU; reruns reproduced the outputs (byte-identical fits). Code and data are listed under Data availability at the end of the paper.

5.2. **Validation of T against the zeros.** Using the first 10^5 zeros of ζ from Odlyzko's tables [10] (with RH assumed for these zeros, which is verified numerically at this height), we compared $T(N)$ from (5.1) with the double series (1.1) truncated to the first K zeros, $T_K(N)$. As a preliminary check, $A(10) = -0.207974$, $A(100) = 13.203470$ and $A(500) = -111.013359$ by the explicit formula, against -0.207974, 13.203468 and -111.013399 from 10^5 zeros. The values $N = 58$ and 74 were chosen with $P(N)^2$ large. Table 2 and Figure 1 show that T_K converges to $T(N)$, while the alternatives "T already contains $\mp P^2$" lie at distance about 15 and are excluded. At the prime power $N = 25 = 5^2$ the truncated series stabilises at an offset of about -0.65 from $T(N)$; we do not analyse this offset here. It is $O(\log^2 N)$ in size and irrelevant for $F(N)/N$.

TABLE 2. Representation (5.1) versus the truncated double series.

N	$T(N)$	$P(N)^2$	$T-2P^2$	$T+2P^2$	T_{1000}	T_{4000}	T_{16000}	T_{40000}
58	12.3251	7.3257	−2.3263	26.9764	12.0896	11.8481	12.2736	12.3414
74	−41.2733	8.6016	−58.4765	−24.0701	−42.7178	−41.5804	−41.4845	−41.2693
25	7.3799	0.0018	7.3762	7.3836	6.8567	6.7319	6.7592	6.7247

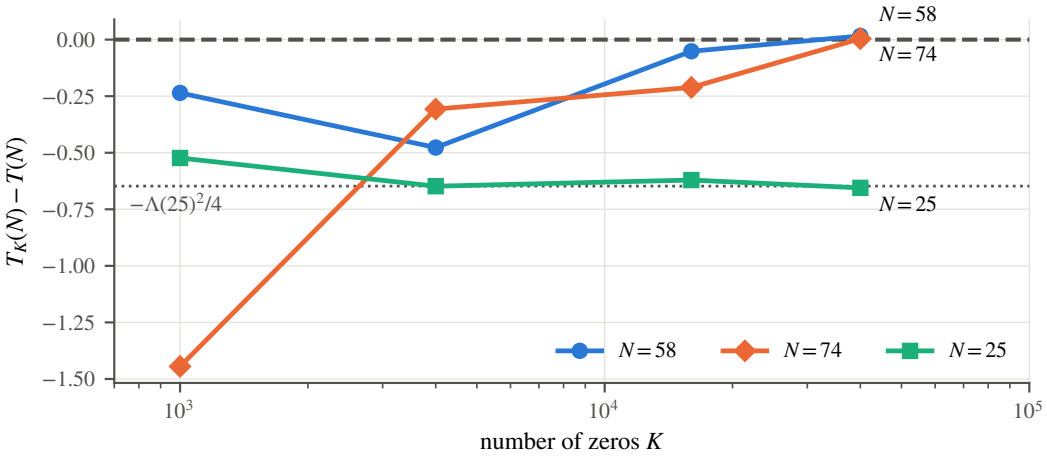


FIGURE 1. $T_K(N)-T(N)$ against the number K of zeros. For $N=58$ and 74 the difference tends to 0 (+0.016 and +0.004 at $K=4\cdot10^4$); the hypotheses $T\mp2P^2$ would sit near ∓14.7 and ∓17.2, outside the frame. For $N=25=5^2$ the difference settles near −0.65 (see the text below Table 2); dotted line: the level $-\Lambda(25)^2/4$, for reference only.

5.3. **Fits of F.** Table 3 summarises the least-squares fits $F(N)/N=c+\beta P(N)^2/N$. Figure 2 shows the data against the predicted line, and Figure 3 the residual $(F(N)-P(N)^2)/N$, which is constant to within 10^{-4} and shows no $N\log\log N$ signature at values of N with $2N$ highly composite.

TABLE 3. Fits of $F(N)/N$ in the two implementations. Last data row: $2N$ of primorial type (A) and the N with $r_G(2N)/2N>3$ (B), mean ± s.d. of $(F-P^2)/N$.

	A (1560 N, $[10^6,5\cdot10^6]$)	B (409 N, $[2.2\cdot10^5,6.0\cdot10^6]$)
fit $F/N=c+\beta P^2/N$	$c=-7.351498$, $\beta=0.999933$	$c=-7.351500$, $\beta=0.999875$
s.d. of residual	$6.66\cdot10^{-5}$	$7.55\cdot10^{-5}$
$(F-P^2)/N$: mean (s.d.)	-7.351501 ($6.67\cdot10^{-5}$)	-7.351506 ($7.58\cdot10^{-5}$)
$(F-P^2)/N$: range	$[-7.351697,-7.351278]$	$[-7.351816,-7.351116]$
s.d. of $(F-kP^2)/N$, $k=0,1,2$	$6.344\cdot10^{-2}$ ($k=0$)	$5.692\cdot10^{-2}$, $\mathbf{7.581\cdot10^{-5}}$, $5.694\cdot10^{-2}$
max P^2/N	0.3553	0.3006
corr$\big((F-P^2)/N,\ r_G(2N)/2N\big)$	−0.0403	0.011
$2N$ highly composite	60 N: $-7.351498\pm9.98\cdot10^{-5}$	83 N: $-7.351504\pm7.42\cdot10^{-5}$
$-4\log2\pi$	-7.351508	

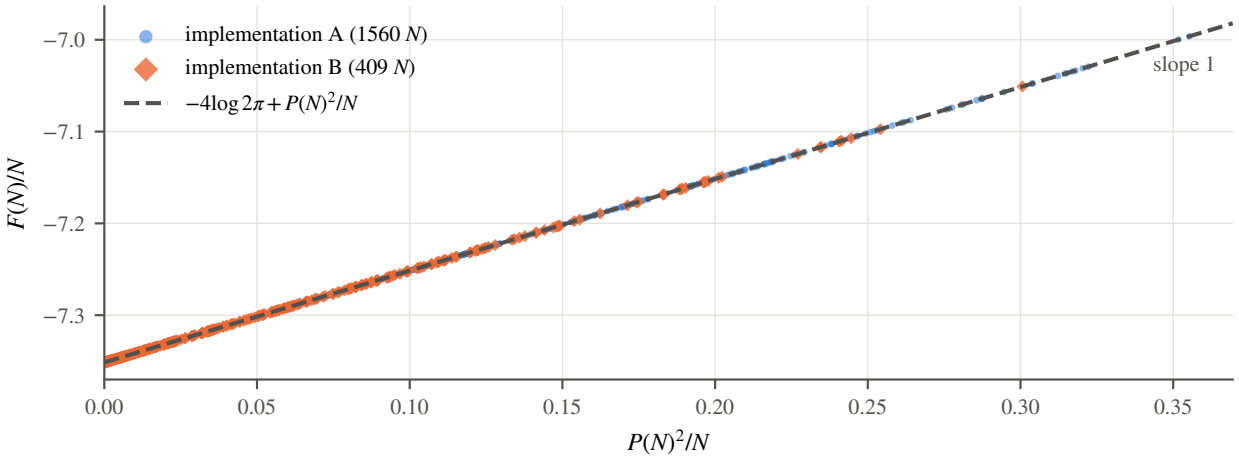


FIGURE 2. $F(N)/N$ against $P(N)^2/N$ for the 1969 computed values of N. Dashed line: slope 1, intercept $-4\log 2\pi$, as predicted by Theorem 1.1 and Remark 2.1.

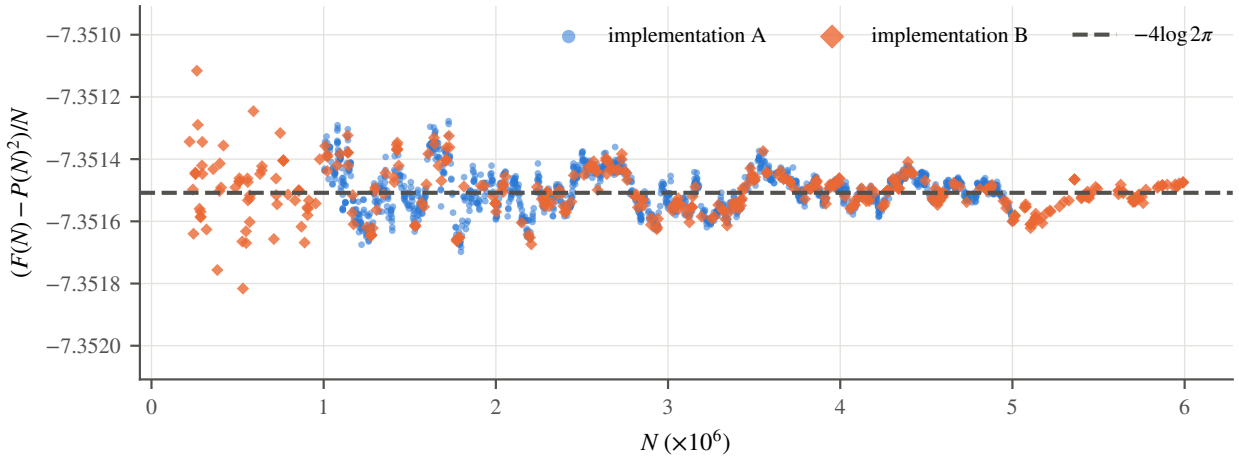


FIGURE 3. The residual $(F(N) - P(N)^2)/N$ against N, in both implementations. Dashed line: $-4\log 2\pi$.

As an example, at $2N = 2012010$ (implementation A), where $r_G(2N)/2N = 5.188530$, one has $(F - P^2)/N = -7.351437$, and the same value at $N \pm 1$ (-7.351437, -7.351431), while $T(N)/N$ goes from -2.212 to $+0.296$ to $+2.810$: the jumps of r_G are carried entirely by T, as Theorem 1.3 predicts.

The data show that F contains exactly one copy of P^2: $\beta = 1.0000 \pm 10^{-4}$ in both implementations, and $k = 1$ is the only coefficient for which $(F - kP^2)/N$ has standard deviation of order 10^{-4}. They do *not* show $F \neq O(N)$, since $P(N)^2/N \leq 0.36$ in this range; that is an asymptotic statement, which follows only from Littlewood's theorem.

6. The case $k = 0$

Cantarini notes ([3], p. 3) that the first part of $T(N)$ is "what we expect considering the formula in [BKP] and taking $k = 0$". Brüdern, Kaczorowski and Perelli [2] prove explicit formulae for the Cesàro–Riesz means of order $k > 0$ and call their formula for $k = 0$ purely formal; Theorem 1.3 is consistent with this, since for $k = 0$ the term over pairs of zeros must carry the jumps $r_G(n)$ and cannot be $O(N)$.

Data availability

Code and data are available at `https://github.com/giovanedms/goldbach-cantarini` (Zenodo DOI to be assigned).

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Declaration of competing interest

The author declares no competing interests.

References

1. Gautami Bhowmik and Jan-Christoph Schlage-Puchta, *Mean representation number of integers as the sum of primes*, Nagoya Math. J. **200** (2010), 27–33, arXiv:0806.3295.
2. Jörg Brüdern, Jerzy Kaczorowski, and Alberto Perelli, *Explicit formulae for averages of Goldbach representations*, Trans. Amer. Math. Soc. **372** (2019), no. 10, 6981–6999, arXiv:1712.00737.
3. Marco Cantarini, *Explicit formula for the average of Goldbach numbers*, Indian J. Math. **61** (2019), no. 2, 253–279, Zbl 1425.11165. Preprint version: *Explicit formula for the average of Goldbach and prime tuples representations*, arXiv:1801.08475v1 (2018); all equation and page numbers in the present paper refer to the arXiv version.
4. Akio Fujii, *An additive problem of prime numbers*, Acta Arith. **58** (1991), 173–179.
5. Patrick X. Gallagher, *A large sieve density estimate near* $\sigma = 1$, Invent. Math. **11** (1970), 329–339.
6. Daniel A. Goldston and Ade Irma Suriajaya, *On an average Goldbach representation formula of Fujii*, Nagoya Math. J. **250** (2023), 511–532, arXiv:2110.14250.
7. Albert E. Ingham, *The distribution of prime numbers*, Cambridge Tracts in Mathematics, vol. 30, Cambridge University Press, 1932, Reprinted by Stechert-Hafner, New York, 1964, and Cambridge University Press, 1990.
8. John E. Littlewood, *Sur la distribution des nombres premiers*, C. R. Acad. Sci. Paris **158** (1914), 1869–1872.
9. Hugh L. Montgomery and Robert C. Vaughan, *Multiplicative number theory I. Classical theory*, Cambridge Studies in Advanced Mathematics, vol. 97, Cambridge University Press, 2007.
10. Andrew Odlyzko, *Tables of zeros of the Riemann zeta function: the first 100,000 zeros*, `www-users.cse.umn.edu/~odlyzko/zeta_tables/`.
11. Olivier Ramaré and Yannick Saouter, *Short effective intervals containing primes*, J. Number Theory **98** (2003), no. 1, 10–33.

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